SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Case Study **Duration: 2hrs**

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior.**(5 Marks)**
2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation.**(5 Marks)**
3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**

5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**

6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**

7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.**(5 Marks)**

8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**

9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations.**(5 Marks)**

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

*I **K Sumanth Kumar Reddy (192425222)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sumanth

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q1. RL-Based Warehouse Robot (Pick, Pack, Transport)

Understanding of the Case

The logistics company's warehouse robot must operate inside a large, dynamic facility, continuously picking items from shelves, packing them at designated stations, and transporting completed orders to dispatch zones. The environment is shared with human workers, other autonomous robots, and moving material-handling equipment, so the robot cannot rely on a fixed, pre-planned route. The core challenge is therefore to learn a sequential decision-making policy through trial and error that jointly optimizes three often competing goals — completing orders quickly, picking and packing the correct items with high accuracy, and avoiding any collision or safety hazard. This makes the problem a natural fit for reinforcement learning, since the robot must learn from repeated interaction with the warehouse rather than from a single static plan, and must adapt as shelf layouts, order volumes, and congestion patterns change over time.

MDP Formulation

Environment: The warehouse floor is represented as a grid or graph in which nodes correspond to aisles, shelf locations, packing stations, and charging docks. The environment includes static obstacles such as racks and pillars as well as dynamic obstacles such as human workers and other robots, and it evolves continuously as new orders arrive and inventory levels change.

Reward Function Design

A balanced reward function can be expressed as $R = w1 \cdot (\text{speed term}) + w2 \cdot (\text{accuracy term}) + w3 \cdot (\text{safety term})$, with the weights tuned according to operational priorities such as service-level agreements and safety regulations.

Analysis & Evaluation of Reward Design

Reward design critically shapes the robot's emergent behaviour, and the three components must be kept in balance. If the speed term dominates, the robot may learn to take risky shortcuts through congested aisles, increasing collisions — a classic case of reward hacking, where the agent maximizes the numeric reward while violating the designer's true intent. Conversely, if the safety penalty dominates excessively, the robot becomes overly cautious, waiting or re-routing so often that throughput collapses and delivery deadlines are missed. Using only a single sparse reward granted at final delivery would also slow learning considerably, since the robot would receive no feedback across the many intermediate pick and pack decisions that make up an episode; dense, shaped rewards at each sub-task resolve this. In practice, the safety penalty should be calibrated so that no combination of speed or accuracy gain can ever make an unsafe shortcut profitable in expectation, guaranteeing that the learned policy remains reliable, efficient, and safe when deployed in a real warehouse alongside human workers.

Conclusion

Overall, treating the warehouse robot's task as an MDP with a carefully weighted, multi-component reward gives the logistics company a principled way to balance throughput, accuracy, and safety, and provides a clear framework for iteratively re-tuning behaviour as warehouse layouts, order volumes, or safety standards evolve.

Q2. RL-Based Dynamic Movie Recommendation

Understanding of the Case

The streaming platform must recommend content that keeps users engaged over time, adapting continuously as tastes shift and new titles are added to the catalogue. Unlike a static, one-shot recommendation, reinforcement learning treats recommendation as a sequential decision process in which each suggestion shapes the user's future behaviour, mood, and trust in the platform, not merely an isolated click. This framing allows the system to reason about long-term satisfaction and retention rather than optimizing only for the next click, which is essential in a competitive market where users can easily switch to another service if recommendations feel repetitive or manipulative.

MDP Formulation

Environment: The streaming platform itself, encompassing its catalogue, user interface, and the broader ecosystem of competing entertainment options that influence whether a user continues watching or leaves the session.

Reward Signal Analysis

The choice of reward signal has an outsized effect on platform health. If reward is based only on clicks, the system learns to favour sensational thumbnails and titles that generate a click but do not sustain genuine engagement, degrading trust over time. Rewarding watch-time and completion instead better reflects true viewer satisfaction, since it requires the content to actually hold attention. Incorporating long-term retention or subscription renewal as a delayed reward further discourages short-term manipulative recommendations, but it introduces a harder credit-assignment challenge, because the influence of any single recommendation on a renewal decision weeks later is difficult to isolate from the many other recommendations shown in between.

Exploration vs Exploitation Trade-off

Exploitation recommends titles closely similar to a user's known preferences, maximizing short-term engagement, but risks creating a narrow filter bubble that reduces long-term discovery, catalogue utilization, and eventual satisfaction. Exploration — implemented through mechanisms such as epsilon-greedy selection, Thompson sampling, or upper-confidence-bound bandits — occasionally surfaces novel genres or lesser-known titles to learn about broader latent preferences, at the cost of some short-term engagement if the exploratory pick does not resonate. A well-tuned balance, for example heavier exploitation for established loyal users and more active exploration for new or ambiguous users, tends to maximize long-run engagement, catalogue-wide discovery, and platform stickiness simultaneously.

Conclusion

In summary, framing recommendation as an RL problem lets the platform optimize for genuine, sustained viewer satisfaction rather than momentary clicks, provided the reward signal and exploration strategy are deliberately designed to avoid short-termism and filter bubbles alike.

Q3. RL for Autonomous Irrigation and Fertilizer Optimization

Understanding of the Case

The agricultural system must decide how much water and fertilizer to apply to a field based on continuously evolving soil conditions and weather forecasts, where the ultimate measure of success — crop yield at harvest — is only known long after each individual irrigation or fertilization decision is taken. This delayed, cumulative outcome, combined with the physical cost of experimenting on a live crop, makes the problem considerably harder than typical RL benchmarks, and requires a framework that can learn efficiently from a limited number of growing seasons while still responding sensibly to day-to-day fluctuations in weather and soil readings.

MDP Formulation

Environment: The field itself, modelled through sensor networks and weather-forecast feeds, in which soil and crop conditions change slowly compared to a single action but respond cumulatively to a whole season of decisions.

Effect of Delayed Rewards

Since the true reward — yield — is only observed at harvest, weeks or months after each irrigation decision, the agent faces a severe credit-assignment problem: it is difficult to determine which specific historical actions were responsible for the final outcome, especially when weather introduces additional confounding variation. This is mitigated by using intermediate proxy rewards such as growth-rate or chlorophyll-index improvements measured throughout the season, by applying reward shaping to provide more frequent feedback, and by tuning the discount factor so that early, well-timed irrigation or fertilization decisions are not unfairly undervalued relative to actions taken closer to harvest.

Algorithm Justification

Because the irrigation and fertilizer actions are naturally continuous quantities and the reward signal is both delayed and comparatively sparse, actor-critic methods such as Proximal Policy Optimization (PPO) or Deep Deterministic Policy Gradient (DDPG) are well suited to this task, since they handle continuous action spaces directly and use a learned value/critic estimate to propagate delayed reward information back to earlier states far more efficiently than pure Monte Carlo return estimation. Given the high real-world cost and slow feedback loop of experimenting directly on a live crop, it is strongly recommended to first train the policy extensively in a simulated digital-twin environment calibrated against historical agronomic data, before fine-tuning cautiously on the real field.

Conclusion

In conclusion, an RL-based irrigation and fertilization system, supported by proxy rewards and simulation-first training, can learn resource-efficient policies that improve yield over successive growing seasons while respecting the practical constraints of delayed feedback and costly real-world experimentation.

Q4. RL for Real-Time Network Intrusion Detection

Understanding of the Case

A cybersecurity reinforcement-learning agent must continuously monitor high-volume network traffic and decide, in real time, how to respond to potential attacks as they occur. The problem is uniquely challenging because genuine attacks form only a tiny fraction of all observed traffic, the adversary actively adapts its behaviour to evade detection, and any response action must be taken within a strict latency budget to be useful, since a delayed reaction to an active attack can allow significant damage before the system intervenes.

State and Environment Modelling

State (S): Statistical traffic features such as packet rate, protocol distribution, and connection duration, alongside system and application logs, flagged anomaly scores from lightweight detectors, and short-term traffic history that captures recent trends.

Environment: A non-stationary network in which both legitimate traffic patterns (driven by business cycles, user behaviour, and new applications) and attacker strategies continuously evolve, requiring the agent's policy to adapt continually rather than rely on a single fixed model trained once and never updated.

Action and Reward Design

Action (A): Allow the traffic to pass, block or blacklist a source IP, isolate a compromised host from the rest of the network, throttle suspicious bandwidth, or raise an alert for human security-analyst review rather than acting autonomously.

Policy (π): A decision-making model that continuously scores incoming traffic and selects the response action expected to minimize the combined cost of missed attacks, false alarms, and response delay.

Reward (R): A positive reward for correctly identifying and neutralizing an attack; a negative reward for false positives that block legitimate traffic and hurt usability; a substantially larger negative reward for false negatives, i.e., missed attacks; and an additional penalty proportional to detection latency to discourage slow responses.

Challenges: Sparse Rewards & Real-Time Decisions

Because actual attacks form a tiny fraction of overall traffic, the reward signal is inherently sparse, slowing learning if the agent must wait for rare true-attack episodes; this is addressed with auxiliary, anomaly-based shaped rewards and by injecting simulated attacks during training. Real-time constraints demand low-latency inference, limiting model complexity and requiring efficient function approximators. The reward function must also balance detection accuracy against response speed, and remain cost-sensitive to reflect the asymmetric risk of a missed attack versus an occasional false alarm.

Conclusion

Overall, an RL-based intrusion-detection agent can offer faster, more adaptive protection than static rule-based systems, provided its reward design and training pipeline specifically compensate for the sparsity of true attacks and the strict real-time latency demands of live network defence.

Q5. RL for Ride-Sharing Driver Allocation and Dynamic Pricing

Understanding of the Case

The ride-sharing platform must simultaneously decide where to position idle drivers and how to price rides in each zone under constantly shifting, uncertain demand, all while balancing business revenue, driver earnings, and rider satisfaction. This is a joint sequential-decision problem because a positioning or pricing decision made now affects driver availability and rider willingness to book several minutes later, so a myopic, one-step-at-a-time approach performs poorly compared with a policy that reasons about the evolving city-wide state over time.

MDP Formulation

Environment: The city, discretized into zones, in which demand and traffic evolve continuously and are further disrupted by weather, events, and time-of-day commuting patterns.

Effect of Environmental Uncertainty

Demand is highly stochastic and influenced by unpredictable events such as sudden weather changes, concerts, sporting events, or accidents that create localized surges, so the learned policy must generalize and adapt online rather than memorize fixed historical patterns that may not repeat. Techniques such as distributional reinforcement learning, which models a range of possible outcomes rather than a single expected value, or ensemble and robust-policy methods, help the agent account for this variance and avoid brittle decisions that fail badly under unusual or extreme demand surges that were rarely seen during training.

Ethical Evaluation

Dynamic or surge pricing raises significant fairness concerns, particularly during emergencies or crises where sharply higher prices can be viewed as exploitative rather than a legitimate response to genuine scarcity of drivers. Geographic pricing patterns may also disproportionately affect lower-income neighbourhoods if algorithmic optimization is left entirely unconstrained. Responsible deployment therefore requires explicit surge-price caps, clear transparency to riders about how and why prices are set, and ongoing monitoring for discriminatory or exploitative outcomes across different regions, times, and demographic groups, alongside mechanisms for regulatory and public accountability.

Conclusion

In summary, RL offers ride-sharing platforms a powerful way to jointly optimize driver allocation and pricing under uncertainty, but its deployment must be paired with explicit fairness constraints so that algorithmic efficiency gains do not come at the expense of equitable treatment of riders and drivers.

Q6. RL for Smart Home Automation (Lighting, Heating, Cooling)

Understanding of the Case

The smart home system must control heating, cooling, and lighting so as to keep occupants comfortable while minimizing energy consumption, using occupancy sensing and weather forecasts as key inputs. Because occupant comfort needs and energy prices vary throughout the day and across seasons, the system must continuously adapt its control strategy rather than rely on fixed schedules, making it a natural sequential decision-making problem where today's heating decision affects tomorrow's energy cost and comfort trajectory.

MDP Formulation

Environment: The physical house, including its thermal dynamics, sensor network, and connection to the electricity grid and its time-varying pricing.

Handling Conflicting Objectives

Comfort and energy-saving are directly competing objectives, since the most comfortable setting is rarely the most energy-efficient one. This conflict is commonly handled with a weighted multi-objective reward of the form $R = \alpha \cdot \text{comfort} - \beta \cdot \text{energy\_cost}$, or with a Pareto/constrained-optimization formulation in which energy use is minimized subject to a hard comfort constraint (for example, temperature must stay within a specified band). Either approach allows designers to tune the trade-off explicitly according to occupant tolerance, rather than leaving the balance to an implicit and potentially unpredictable emergent behaviour.

Impact of Reward Shaping

Reward shaping — for example adding intermediate signals for accurate occupancy prediction, or a bonus for pre-emptive pre-cooling before an expected hot afternoon — can speed up learning considerably and produce smoother, more proactive control rather than reactive, oscillating adjustments. However, poorly tuned shaping terms can inadvertently bias the agent toward over-prioritizing energy savings at the expense of comfort, or vice-versa, if the shaping reward is not carefully weighted relative to the primary objectives. Shaping terms should therefore always be validated against real user satisfaction feedback collected over time, rather than against energy metrics alone, to ensure the household actually experiences the intended balance.

Conclusion

Overall, a well-designed multi-objective RL framework allows a smart home to learn genuinely personalized comfort-energy trade-offs over time, turning a fixed, one-size-fits-all thermostat schedule into an adaptive system that reflects each household's actual preferences and habits.

Q7. RL Agent for a Complex Strategy Game

Understanding of the Case

The gaming company needs an agent that can plan and act over long time horizons against skilled human opponents, where individual moves only pay off through their eventual contribution to winning the game, often dozens or hundreds of moves later. This long-horizon credit-assignment problem, combined with a huge combinatorial space of possible game states and an adversary whose strategy is itself evolving, makes complex strategy games one of the most demanding classical testbeds for reinforcement learning, requiring both deep strategic planning and robust generalization against varied opponents.

MDP Formulation

Environment: The game itself, including its rules engine and, critically, the opposing player, whose strategy effectively makes the environment adversarial and non-stationary from the agent's perspective.

Reward Structure for Long-Term Strategy

To encourage long-term strategic play rather than short-sighted tactics, potential-based reward shaping is used, since it can be shown mathematically to preserve the optimal policy while still providing much denser feedback throughout a long game. Rewards for intermediate milestones such as map control or economic advantage are scaled modestly relative to the terminal win/loss signal, ensuring the agent never learns to sacrifice an eventually winning position purely to accumulate more intermediate reward along the way.

Evaluation of Exploration Strategies

Exploration mechanisms such as self-play combined with Monte Carlo Tree Search, entropy bonuses that encourage varied move selection, or curiosity-driven exploration that rewards visiting novel game states, all allow the agent to discover creative strategies rather than converging prematurely to a single, narrow, and ultimately exploitable style of play. Too little exploration leads to predictable, easily-countered strategies once human opponents learn the agent's patterns; too much exploration slows convergence and wastes training time on unproductive lines of play, so exploration is typically annealed — starting high early in training and gradually reduced — as the policy matures and self-play opponents become stronger.

Conclusion

In summary, modelling the game as an MDP with shaped, potential-based rewards and a carefully annealed exploration schedule enables the agent to develop genuinely long-term strategic play that generalizes well against varied and skilled human opponents, rather than a brittle set of memorized tactics.

Q8. RL for Real-Time Fraudulent Transaction Detection

Understanding of the Case

The financial institution needs to flag fraudulent transactions instantly, at the moment a card or account is used, while minimizing disruption to the overwhelming majority of legitimate customers. The task is made especially difficult by extreme class imbalance, since genuine fraud represents only a tiny fraction of all transactions, and by delayed ground-truth labels, since confirmation of fraud (through customer complaints or chargebacks) often arrives long after the transaction itself was processed, complicating the feedback the learning system can use to improve.

MDP Formulation

Environment: The payments processing system, through which a continuous, high-volume stream of transactions flows, each requiring a near-instant decision.

Challenges: Imbalanced Data & Delayed Feedback

Because fraudulent transactions form a very small fraction of all transactions, a naive agent can achieve seemingly high accuracy by almost always approving, so reward weighting or explicitly cost-sensitive learning is required to counter this imbalance and force the agent to pay meaningful attention to the rare fraud cases. Ground-truth confirmation of fraud, for instance through a customer-reported chargeback, often arrives days or weeks after the original transaction, creating a delayed-feedback problem that requires techniques such as eligibility traces to link the delayed signal back to the responsible decision, along with periodic offline relabelling of historical transactions and scheduled model retraining as confirmed labels accumulate.

Risk Evaluation

Incorrect decisions carry clearly asymmetric risk: false negatives cause direct financial loss to the institution and its customers along with potential regulatory exposure, while excessive false positives damage customer trust, generate support costs, and can drive customers to competitors. The reward function must therefore be carefully calibrated to reflect these differing real-world costs rather than treating both error types equally, and human oversight should remain firmly in the loop for high-value, borderline, or otherwise ambiguous transactions where an automated decision alone carries too much risk.

Conclusion

In conclusion, an RL-based fraud detection system can adapt faster than static rule engines to evolving fraud patterns, but its success depends on cost-sensitive reward design, mechanisms to handle delayed labels, and continued human oversight for the highest-risk decisions.

Q9. RL for Personalized Rehabilitation Exercise Recommendation

Understanding of the Case

The healthcare system must recommend rehabilitation exercises tailored to each patient's evolving recovery progress, where patients vary widely in age, condition severity, and physiological response, and where an unsafe or poorly timed recommendation could cause real physical harm rather than merely a poor outcome. This combination of high inter-patient variability and genuine safety stakes distinguishes this application from most other RL use cases and demands a cautious, personalized approach to both learning and deployment.

MDP Formulation

Environment: The patient's own recovering body, monitored indirectly through wearable sensors, self-reports, and periodic clinical assessment, all of which evolve slowly and noisily over the course of rehabilitation.

Delayed Rewards and Patient Variability

True recovery outcomes are only evident over multiple weeks, creating a delayed and inherently noisy reward signal, further complicated by wide variability across patients in age, condition severity, and comorbidities that make a single global response pattern unreliable. This is addressed through personalization — treating each patient's history contextually, for example via contextual bandits or hierarchical reinforcement learning that adapts a shared base policy to the individual — and by using shorter-term proxy indicators of progress, such as small week-to-week mobility gains, to provide more frequent and informative feedback than waiting for the full recovery outcome alone.

Ethical and Safety Evaluation

Because an incorrect recommendation can physically harm an already vulnerable patient, this is a genuinely safety-critical application requiring conservative, constrained exploration — often termed safe reinforcement learning — rather than the free, unconstrained trial-and-error used in lower-stakes domains. Mandatory human-clinician oversight should be built in for any escalation or unusual case, and the system must ensure strict protection of sensitive health data together with informed patient consent before any automated recommendation is acted upon, reflecting both ethical obligation and regulatory requirement in healthcare settings.

Conclusion

Overall, a carefully constrained, personalized RL framework can meaningfully improve rehabilitation outcomes by adapting to each patient's unique recovery trajectory, provided safety, clinician oversight, and data privacy remain the top design priorities throughout.

Q10. RL for Smart Grid Electricity Distribution

Understanding of the Case

The smart grid system must balance electricity supply and demand in real time, reducing peak load while maintaining uninterrupted reliability, across a network composed of many interacting generation sources, storage units, and consumers spread over a wide geographic area. Unlike most other RL applications discussed, failures here have immediate, large-scale physical consequences, since even a brief reliability lapse can cascade into a regional blackout, making both the design of the reward function and the deployment strategy exceptionally conservative compared with lower-stakes domains.

MDP Formulation

Environment: The electricity grid itself, spanning multiple regions, generation types, and storage assets, all coupled together and subject to real-time physical constraints on power flow and stability.

Reward Function for Efficiency vs Reliability

The reward function must weight reliability far more heavily than pure efficiency, since a blackout carries severe real-world consequences for hospitals, industry, and households alike. This can be expressed as $R = \text{efficiency\_term} - \lambda \cdot \text{peak\_penalty} - \mu \cdot \text{reliability\_violation}$, with the coefficient μ set high enough that the agent can never rationally trade away reliability for only a marginal gain in efficiency, ensuring that cost-saving behaviour is always subordinate to keeping the grid stable.

Challenges: Scalability & Real-Time Learning

The grid's state and action space grows rapidly with the number of nodes and substations involved, making a single centralized policy computationally infeasible at national or even regional scale; this motivates decentralized or multi-agent reinforcement learning combined with function approximation to keep the problem tractable. Real-time learning is further constrained by the grid's safety-critical nature, so training is typically performed extensively in high-fidelity simulated digital-twin environments first, with only conservative, thoroughly verified policies deployed to the real grid, where they are then continuously fine-tuned online but always kept within strictly enforced safe operating limits.

Conclusion

In summary, reinforcement learning offers smart grids a path toward more efficient, adaptive electricity distribution, but realizing this in practice requires decentralized architectures for scalability and a reward structure that never compromises reliability for marginal efficiency gains.